FL-02 – Portfolio Case Studies

Name: Ahmed Sherif
Role: AI/ML Engineer | Data Scientist

Voice Card

Technical, direct, honest, practical, no buzzwords.

Case Study 1 – HouSmart AI Platform

Problem

The project required building scalable AI-powered backend services capable of supporting real estate intelligence, including property scoring, geographic data processing, and policy-aware recommendations. The solution needed to be maintainable, production-ready, and easy to extend.

What I Did

I designed and implemented backend services using FastAPI and PostgreSQL/PostGIS while integrating AI workflows into the platform. I built data-processing components, implemented APIs, improved data validation, and worked on services that transformed raw geographic and demographic data into structured information. Throughout development, I focused on writing clean, maintainable code and making architectural decisions that would support future growth.

Outcome

The result was a more reliable and scalable backend capable of supporting AI-driven features. The modular architecture made it easier to extend the platform with additional services while maintaining code quality and development speed.

Case Study 2 – PersonaLLM

Problem

I wanted to build an AI assistant capable of generating personalized responses based on a user's own knowledge and context instead of producing generic answers.

What I Did

I designed the system architecture, integrated LLM technologies, implemented Retrieval-Augmented Generation (RAG), managed vector embeddings, and connected the retrieval pipeline with language models. I also organized the project for maintainability and future improvements.

Outcome

The project demonstrated my ability to combine modern AI technologies into a complete application rather than working with individual models in isolation. It strengthened my understanding of LLM application development and AI system design.

Case Study 3 – Real-Time ML System

Problem

Many machine learning projects only work on static datasets, while production systems require continuous data ingestion and real-time prediction.

What I Did

I built a real-time machine learning pipeline that collected streaming cryptocurrency data, processed technical indicators, incorporated market sentiment, and generated predictions through a modular architecture. I focused on automation, scalability, and clean deployment practices.

Outcome

The project demonstrated end-to-end machine learning engineering skills, including data pipelines, feature engineering, model integration, and production-oriented system design.

About

I'm Ahmed Sherif, an AI/ML Engineer passionate about building production-ready AI systems that solve real business problems. I enjoy combining machine learning, backend engineering, cloud technologies, and modern AI tools to create reliable, scalable applications.

Contact / Call to Action

If you're looking for an AI/ML Engineer who can build production-ready AI systems and contribute to real-world engineering challenges, I'd be happy to discuss how I can help. Please contact me to schedule an interview or connect about opportunities.

Before vs. After

Generic AI Version

"I am a results-driven AI engineer passionate about leveraging cutting-edge technologies to deliver innovative solutions and maximize business value."

My Edited Version

"I build production-ready AI systems that solve real business problems. I focus on clean engineering, scalable architectures, and practical solutions that can be deployed and maintained."

Why This Version Is Better

The edited version reflects how I actually describe my work. It avoids generic phrases like "results-driven" and "cutting-edge," focuses on specific engineering strengths, and directly supports my portfolio's claim for AI Engineering hiring managers.